



Department of Information Technology

Academic Year 2025 – 2026 (Odd Semester)

Degree, Semester & Branch: V Semester B.Tech. IT

Course Code & Title: CS3551 & Distributed Computing

Name of the Faculty member (s): Mrs. A. Alagulakshmi, AP/IT

Innovative Practice Description

- **Unit / Topic:** Unit IV / Check pointing and Rollback Recovery
- **Course Outcome:** CO4
- **Activity Chosen:** Think-Pair- Share

- **Justification:**

The topic Checkpointing and Rollback Recovery involves understanding complex mechanisms used to achieve fault tolerance in distributed systems. To simplify and deepen learning, the Think–Pair–Share activity is chosen. In this method, students first think individually about how checkpointing helps recover from system failures, then pair with a classmate to discuss their ideas, compare approaches, and clarify doubts. Finally, they share their conclusions with the larger group, promoting collaborative learning and critical thinking. This activity encourages students to analyse real-world scenarios where checkpointing is applied, such as in databases and cloud systems. It also enhances conceptual clarity, teamwork, and communication skills while reinforcing the principles of recovery and reliability in distributed computing.

- **Time Allotted for the Activity:** 20 Minutes

- **Details of the Implementation:**

Instructor Introduction: The instructor explains the concept of Checkpointing and Rollback Recovery, highlighting its importance in achieving fault tolerance in distributed systems. Students are asked to review related materials and examples before class.

Think (5 Minutes): Students individually think about how a system can recover from process failure using checkpoints and how rollback helps restore system consistency.

Pair (5 Minutes): Students pair up to discuss their individual ideas, compare different checkpointing strategies (such as coordinated and uncoordinated checkpointing), and identify challenges like domino effects and message logging.

Share (Class Discussion): Each pair presents their discussion outcomes to the class. The instructor summarizes key points, clarifies misconceptions, and demonstrates how checkpointing and rollback recovery work in practical distributed environments.

Outcome: This activity helps students strengthen their understanding of fault tolerance mechanisms, develop critical thinking on system recovery design, and enhance peer learning through collaborative discussion and reflection.

• **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO10	PO12	PSO1	PSO2	PSO3
CO2	3	2	1	2	2	2	3	2	2

(1 – Low 2 – Moderate 3 – High)

• **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO4	PO10	PO12	PSO1	PSO2	PSO3
	3	2	1	2	2	2	3	2	2
Justification for correlation	Students apply knowledge of engineering fundamentals and distributed computing principles to understand checkpointing and rollback recovery mechanisms for fault tolerance.	Students identify, formulate, and analyze failures in distributed systems to determine appropriate recovery techniques.	Students design and implement fault-tolerant distributed systems using checkpointing strategies and recovery algorithms.	Students use modern computational tools and simulation environments to model system failures and verify recovery processes.	Students communicate effectively by presenting recovery models and explaining their fault-handling techniques during discussions or reports.	Students understand the impact of reliability and fault tolerance in distributed systems, demonstrating lifelong learning of emerging recovery methods.	Students apply computing concepts to design reliable and efficient distributed systems that can recover from failures.	Students analyze real-world distributed environments to propose effective checkpointing strategies for maintaining system consistency.	Students develop practical fault-tolerant models by integrating rollback recovery techniques into distributed applications.

• **Images / Screenshot of the practice:**



Figure 1: Students discussion with their team members [Jeyalakshmi A, Priyadharshini R, Ashika]

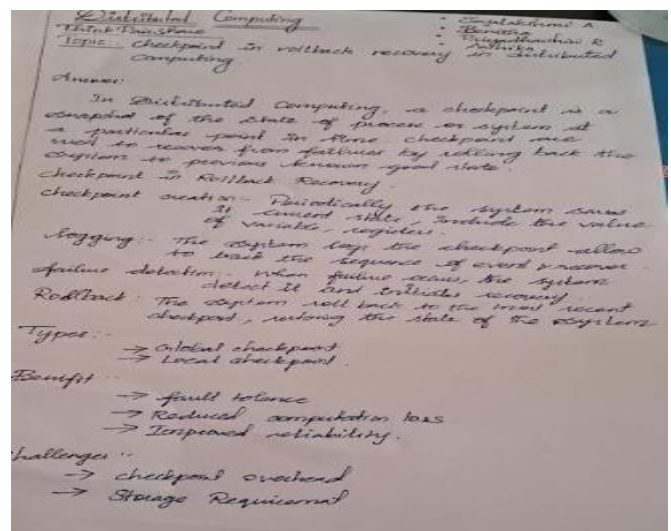


Figure 2: Solved answers shared by the team members [Jeyalakshmi A, Priyadharshini R, Ashika]

Reflective Critique:**❖ *Feedback of practice from students and other stakeholders:***

- All students felt that they engaged fully in the task and group discussions.
- Later, students informed that they explored and shared the concepts of Checkpointing and Rollback Recovery with their peers.
- Students also mentioned that pairing up helped them learn from each other and clarify doubts related to failure recovery mechanisms in distributed systems.

❖ *Benefit of the practice:*

- The activity helped students feel more confident in explaining the Checkpointing and Rollback Recovery concept.
- Through this activity, students gained a clear understanding of how fault tolerance techniques are applied in distributed computing environments.

❖ *Challenges faced in implementation:*

- Some students were less active during discussions, which affected group participation and collaboration.
- A few students initially struggled to explain the rollback and recovery process clearly, leading to minor confusion within teams.

References:

- ❖ <https://www.theteachertoolkit.com/index.php/tool/think-pair-share>
- ❖ https://www.ritrjpm.ac.in/images/computer-science/2021-2022/MGM_VI_Think_Pair_and_Share.pdf

Signature of Faculty Member**HOD**